

Syed Fareed Alam

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EDUCATION

Stevens Institute of Technology, Hoboken, NJ

Master of Engineering, Mechanical Engineering

Expected July 2027

Shaheed Zulfikar Ali Bhutto Institute of Science and Technology, Karachi, Pakistan

Bachelor of Engineering, Mechatronics

Sept 2020 – Jul 2024

GPA: 3.51 | Chancellor Honors Roll | Merit Scholarship

SKILLS

Software: Creo, SOLIDWORKS, Autodesk Inventor, Arduino, Microsoft Office, MATLAB/Simulink

Programming: C++, Python, JavaScript

Certifications: Supervised Machine Learning, Advanced Learning Algorithms

EXPERIENCE

Changan Pakistan, Karachi, Pakistan

Production Trainee Engineer

Nov 2024 – Mar 2025

- Engineered a high-capacity SOLIDWORKS trolley system, increasing intra-line part delivery speed by 15%
- Decreased Defects Per Unit (DPU) by 12% through real-time process refinements and defect data analysis
- Supervised 28 technicians during chassis installation, ensuring 100% adherence to safety and quality through technical guidance

Atlas Honda, Karachi, Pakistan

Engineering Intern

Sept 2024 – Nov 2024

- Designed an automated motorcycle frame conveyor system in SOLIDWORKS to replace manual transport, increasing routing efficiency by 18%
- Designed four ergonomic jigs and fixtures, increasing technicians' productivity by 25% and mitigating injury risks
- Collaborated across assembly, welding, paint, maintenance, and QA teams to streamline cross-functional workflows

Sui Southern Gas Company, Karachi, Pakistan

Manufacturing Intern

Jun 2023 – Sept 2023

- Analyzed G4 and G16 meter production lines, identifying two critical bottlenecks and proposing actionable solutions
- Authored a 12-page process improvement report implemented to reduce production waste
- Supported troubleshooting tasks, ensuring compliance with industrial safety and operating standard

ACADEMIC PROJECTS

Shaheed Zulfikar Ali Bhutto Institute of Science and Technology, Karachi, Pakistan

Desktop Robotic Arm using Computer Vision

Sept 2023 – May 2024

- Reduced production costs by 50% by optimizing and 3D printing robotic arm components in SOLIDWORKS
- Integrated inverse kinematics using Arduino and NEMA 17 stepper motors for high-precision pick-and-place control
- Synchronized a Python computer vision model with C++ hardware control, achieving 90% target accuracy

Open Loop Control Cocktail Machine

Jan 2023 – May 2023

- Designed custom gear systems in SOLIDWORKS for a motorized platform to enhance system reliability
- Improved dispensing accuracy by 80% by programming automated drink modes in C++ and refining timing logic
- Utilized Arduino-based control with precise stepper motor operation to ensure consistent performance

Autonomous Ball Shooting Robot

Jan 2023 – May 2023

- Designed all mechanical components in SOLIDWORKS, ensuring structural integrity for high-speed robotic movements.
- Achieved 90% shooting precision by engineering real-time trajectory adjustments in C++ using ultrasonic sensor feedback.
- Reached the semi-finalist stage in a robotics competition among 15+ teams, recognized for innovative mechanical design